# THE $p$-ADIC HEIGHT PAIRING AT ANOMALOUS PRIMES AND POINTS OF ORDER $p$ OVER $\mathbb{Q}_p$

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Abstract. Let $E/\mathbb{Q}$ be an elliptic curve and let $p \geq 5$ be a prime of good ordinary reduction with $\#\widetilde{E}(\mathbb{F}_p) = p$. At such an anomalous prime the cyclotomic $p$-adic regulator need not be $p$-integral, and the standard index count only gives $v_p(\mathrm{Reg}_p) \geq r-2$. Assume that $p$ does not divide the Tamagawa product and that $E(\mathbb{Q})$ surjects onto $\widetilde{E}(\mathbb{F}_p)$. We prove that all entries of the $p$-adic height matrix lie in $\mathbb{Z}_p$ if and only if $E(\mathbb{Q}_p)$ has a point of order $p$. In rank one, $v_p(\mathrm{Reg}_p) = -1$ when there is no such point. We also determine the height matrix modulo $p$: in the non-split case $p$ times the matrix reduces to a rank-one form whose constant is a Fermat quotient of the $p$-th division polynomial, and in the split case the matrix itself reduces to a form of rank at most two whose diagonal constant is a second-order Fermat quotient. Since CM curves are split at every anomalous prime, the results apply in particular to the anomalous primes excluded in recent work on rank-two CM curves. The arithmetic part of the proofs is formalised in Lean 4.

## 1. Introduction

Let $E/\mathbb{Q}$ be an elliptic curve, $p \geq 5$ a prime of good ordinary reduction, and $\Lambda = E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}$, of rank $r$. The cyclotomic $p$-adic height of Mazur–Tate and Schneider [MT91, Sch85] is a $\mathbb{Q}_p$-valued quadratic form $h_p$ on $\Lambda$, and its discriminant $\mathrm{Reg}_p$ is the regulator in the $p$-adic Birch–Swinnerton-Dyer conjecture [MTT, BMS]. This paper is about the $p$-adic valuation of $\mathrm{Reg}_p$ and of the individual entries of the height matrix.

The prime $p$ is *anomalous* if $p \mid \#\widetilde{E}(\mathbb{F}_p)$ [Maz72]; for $p \geq 7$ this means $\#\widetilde{E}(\mathbb{F}_p) = p$. At an anomalous prime the regulator can fail to be $p$-integral. Negative valuations occur in the tables of Mazur, Stein and Tate [MST, §4], and the elementary index count of Theorem 3.2 explains their size: $v_p(\mathrm{Reg}_p) \geq r - 2$. For this reason anomalous primes are often excluded. Balakrishnan, Çiperiani and Stein work at non-anomalous primes, where the height pairing is integral [BCS], and Banwait excludes anomalous primes in his criterion for the Shafarevich–Tate group of rank-two CM curves [Ban]. We describe what happens at these primes.

We say that $E$ is *split at* $p$ if $E(\mathbb{Q}_p)[p] \neq 0$. When $p$ is anomalous, this holds exactly when the reduction sequence $0 \to E_1(\mathbb{Q}_p) \to E(\mathbb{Q}_p) \to \widetilde{E}(\mathbb{F}_p) \to 0$ splits. Let $c_\ell$ be the Tamagawa numbers, $n = \prod_\ell c_\ell$, and let $(\iota)$ be the hypothesis that $\Lambda \to \widetilde{E}(\mathbb{F}_p)$ is surjective.

**Theorem A.** *Assume $\#\widetilde{E}(\mathbb{F}_p) = p$, $p \nmid n$ and $(\iota)$. Let $M$ be the height matrix of $h_p$ on a basis of $\Lambda$.*

1. *All entries of $M$ lie in $\mathbb{Z}_p$ if and only if $E(\mathbb{Q}_p)[p] \neq 0$.*
2. *If $r = 1$, then $v_p(\mathrm{Reg}_p) \geq 0$ if $E$ is split at $p$, and $v_p(\mathrm{Reg}_p) = -1$ otherwise.*
3. *If $E$ is split at $p$, then $v_p(\mathrm{Reg}_p) \geq \max(r-2, 0)$.*

Part (1) concerns the entries of $M$, and does not follow from any bound on $\det M$; it holds in every rank. The bound on $\det M$ in (3) is new only for $r = 1$, where (2) identifies the curves with non-integral regulator.

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**Theorem B.** *Under the hypotheses of Theorem A let $\chi\colon \Lambda \to \mathbb{F}_p$ be the discrete logarithm of the reduction map.*

(1) *If $E$ is not split at $p$, then $pM$ is integral and $pM \equiv 2\lambda_1\, \chi\chi^{t} \pmod p$ with $\lambda_1 \in \mathbb{F}_p^{\times}$; in particular $pM \bmod p$ has rank one.*
(2) *If $E$ is split at $p$, then $M \bmod p$ vanishes on $\ker\chi \times \ker\chi$, so its rank is at most two.*

*In both cases $v_p(\mathrm{Reg}_p) = (r-2) + v_p(\det T)$ for an integral Schur complement $T$.*

Here $\lambda_1$ is half the Fermat quotient of $\phi_p(x(P))$ for a point $P$ whose reduction generates $\widetilde{E}(\mathbb{F}_p)$, where $\phi_p$ is the $p$-th division polynomial. In the split case the diagonal constant is a second-order Fermat quotient of the same polynomial (Proposition 6.7).

**The mechanism.** For $P' \in E(\mathbb{Q})$ whose reduction has order $p$, put

$$\lambda = \tfrac{1}{2}\, q_p\big(\phi_p(x(P'))\big) \in \mathbb{F}_p, \qquad q_p(u) = \frac{u^{p-1}-1}{p} \bmod p.$$

The height of $pP'$ splits into a formal-group term and $2\log_p$ of a ratio of coordinates (Lemma 3.3), and the second $p$-adic digit of the latter is $-\lambda$ (Theorem 4.3). If $E$ is split at $p$, Vélu's factorisation of $[p]$ through a rational subgroup of order $p$ makes $\phi_p(x(T))$ a $p$-th power, so $\lambda = 0$ (Corollary 4.6). Conversely, $\lambda = 0$ implies that $E$ is split at $p$ (Proposition 4.8). In the form "$E$ is split at $p$ if and only if $E \bmod p^2$ is the canonical lift", this equivalence is due to Gross [Gro90] (see [LR, Thm. 4.1]), David–Weston [DW, Lemma 3.2] and Ozeki–Yoshida [OY, Lemma 4.1]. The new ingredient is its formulation through the Fermat quotient of $\phi_p$ at one point, which is what links it to the height. We give a proof in this form in Appendix A.

**Frequency.** David and Weston showed that an elliptic curve over $\mathbb{Q}$ has, on average, only finitely many primes $p$ with $E(\mathbb{Q}_p)[p] \neq 0$ [DW]. For a non-CM curve, therefore, the anomalous primes covered by Theorem A are almost all non-split, and in rank one the regulator has valuation exactly $-1$ there. A CM curve, on the other hand, is split at every anomalous prime of good ordinary reduction [DW, Prop. 2.1(3)]. For the rank-two CM curves of [Ban] this gives Corollary 7.2: at an anomalous prime satisfying our hypotheses the regulator is integral, and it is a unit exactly when $M \bmod p$ has rank two.

**Formalisation.** The arithmetic steps of the proofs are formalised in Lean 4 with Mathlib, with no `sorry` or `native_decide`; the geometric and $p$-adic analytic inputs, which are not available in Mathlib, enter the Lean statements as hypotheses. Appendix B gives the correspondence.

**Earlier version and companion paper.** This paper revises and shortens the first part of a draft that the author posted under the pen name Kiichi [Draft]. The split case is studied further in the companion paper [Comp].

## 2. Notation and normalisation

We fix a global minimal Weierstrass equation for $E$ with coefficients $a_1, \dots, a_6$. For $P \in E(\mathbb{Q}) \smallsetminus \{O\}$ write

$$P = \big(\alpha/d^2,\ \beta/d^3\big), \qquad \alpha, \beta, d \in \mathbb{Z},\ d \geq 1,\ \gcd(\alpha, d) = \gcd(\beta, d) = 1.$$

Let $t = -x/y$ be the parameter of the formal group and $E_k(\mathbb{Q}_p) = \{P : v_p(t(P)) \geq k\}$. The logarithm $\log_p$ is Iwasawa's, with $\log_p p = 0$. The Fermat quotient $q_p\colon \mathbb{Z}_p^{\times} \to \mathbb{F}_p$ is a surjective homomorphism, and

$$\log_p u \equiv -p\, q_p(u) \pmod{p^2}, \qquad v_p(\log_p u) = 1 + v_p(q_p(u)) \qquad (u \in \mathbb{Z}_p^{\times}). \tag{1}$$

**The height.** Let $\sigma_p(t) = t + c_2t^2 + \cdots \in t\mathbb{Z}_p[[t]]$ be the Mazur–Tate sigma function [MT91]; it is odd in the formal logarithm, and $c_2 = a_1/2$. Following [Har, §2.2], for $P$ such that

(A1) $P$ reduces to $O$ modulo $p$, and
(A2) $P$ has non-singular reduction at every bad prime,

put $h_p(P) = 2\log_p(\sigma_p(P)/d(P))$, and extend by $h_p(Q) = h_p(mQ)/m^2$. This is $2p$ times the normalisation of [MST], so the two valuations of $\mathrm{Reg}_p$ differ by $r$. Put $h_p(P,R) = \frac{1}{2}(h_p(P+R) - h_p(P) - h_p(R))$ and $\mathrm{Reg}_p = \det(h_p(P_i,P_j))$ for a basis $(P_i)$ of $\Lambda$.

**Division polynomials.** Write $x(mQ) = \phi_m/\psi_m^2$ and $y(mQ) = \omega_m/\psi_m^3$, with $\phi_m$ monic of degree $m^2$ in $x$. In characteristic $p$ the map $[p]$ factors through Frobenius, so $\phi_p \equiv f(x^p) \pmod p$ for some polynomial $f$.

**Choice of model.** The substitution $(x,y) \mapsto (x, y+sx)$, $s \in \mathbb{Z}$, preserves global minimality, changes $a_1$ to $a_1 + 2s$, and leaves $x$, $\alpha$, $d$, $\phi_p$ and $\psi_p$ unchanged. As $p$ is odd we may take $p \mid a_1$. Heights and their valuations do not depend on this choice; the two summands of Lemma 3.3 do.

**Standing notation.** $\Lambda_0 = \ker(\Lambda \to \widetilde{E}(\mathbb{F}_p))$, $n = \prod_\ell c_\ell$, and $(\iota)$ is the surjectivity of $\Lambda \to \widetilde{E}(\mathbb{F}_p)$. From Section 4 on we assume $\#\widetilde{E}(\mathbb{F}_p) = p$.

## 3. The index bound and a decomposition of the height

**Lemma 3.1.** *Let $P$ satisfy* (A1) *and* (A2)*, and $e = v_p(t(P)) \geq 1$. Then $\sigma_p(P)/d(P) \in \mathbb{Z}_p^\times$, $h_p(P) \in 2p\mathbb{Z}_p$, and $v_p(h_p(P)) = 1 + v_p(q_p(\sigma_p(P)/d(P)))$.*

*Proof.* We have $v_p(\sigma_p(t)) = e$. Since $v_p(x(P)) = -2e$ and $\gcd(\alpha, d) = 1$, $v_p(d) = e$ and $v_p(\alpha) = 0$. The rest is (1). □

**Theorem 3.2.** *Let $k$ be the index in $\Lambda$ of the image of the group of points satisfying* (A1) *and* (A2)*. Then $v_p(\mathrm{Reg}_p) \geq r - 2v_p(k)$. If $p \nmid n$, then $v_p(k) \leq 1$, with equality if and only if the image of $\Lambda$ in $\widetilde{E}(\mathbb{F}_p)$ contains a point of order $p$.*

*Proof.* By Lemma 3.1 the height matrix on a basis of that subgroup has entries in $p\mathbb{Z}_p$, so its determinant has valuation at least $r$; this determinant is $k^2\mathrm{Reg}_p$. The quotient of $\Lambda$ by the subgroup embeds in $\widetilde{E}(\mathbb{F}_p) \times \prod_\ell \Phi_\ell(\mathbb{F}_\ell)$; if $p \nmid n$, its $p$-part comes from $\widetilde{E}(\mathbb{F}_p)$, whose $p$-part is cyclic of order at most $p$ by the Hasse bound. □

This bound is well known in practice (see Section 8); we use it only as a point of comparison.

**Lemma 3.3.** *Let $P$ satisfy* (A1)*,* (A2) *and $e = v_p(t(P))$. Then*

$$h_p(P) = 2\Lambda_\sigma + 2A, \qquad \Lambda_\sigma = \log_p \frac{\sigma_p(t)}{t}, \qquad A = \log_p \frac{\alpha}{\beta} = \sum_{\ell \neq p} (v_\ell(\alpha) - v_\ell(\beta)) \log_p \ell.$$

*Put $b = v_p(\Lambda_\sigma)$ and $c = v_p(A)$. Then $c \geq 1$ and $b \geq \min(e + v_p(a_1), 2e)$. In a model with $p \mid a_1$ this gives $b \geq 2$ for $e = 1$ and $b \geq 3$ for $e \geq 2$.*

*Proof.* Since $t/d = -\alpha/\beta$, we have $\sigma_p(P)/d = (\sigma_p(t)/t) \cdot (-\alpha/\beta)$, and $\log_p(-1) = 0$. By Lemma 3.1, $\alpha$ and $\beta$ are prime to $p$, so $c \geq 1$ by (1). Finally $\sigma_p(t)/t = 1 + \frac{a_1}{2}t + O(t^2)$ with integral coefficients. □

## 4. The criterion

In this section $\#\widetilde{E}(\mathbb{F}_p) = p$. Let $P' \in E(\mathbb{Q})$ satisfy (A2) and have reduction of order $p$, and put $P = pP'$ and $e = v_p(t(P))$.

**Lemma 4.1.** *$E(\mathbb{Q}_p)[p] \neq 0$ if and only if $e \geq 2$.*

*Proof.* Suppose $\tau \in E(\mathbb{Q}_p)$ has order $p$. As $\widehat{E}(p\mathbb{Z}_p)$ is torsion-free for $p \geq 5$, the reduction of $\tau$ is non-zero and generates $\widetilde{E}(\mathbb{F}_p)$. So $R = P' - j\tau \in E_1(\mathbb{Q}_p)$ for some $j$, and $P = pR$. Since $[p](t) = pt + O(t^2)$ with integral coefficients, $v_p(t(pR)) \geq 2$.

Suppose $E(\mathbb{Q}_p)[p] = 0$. Then $E(\mathbb{Q}_p)$ is torsion-free, so $E(\mathbb{Q}_p) \cong \mathbb{Z}_p$ and $E_1(\mathbb{Q}_p) = pE(\mathbb{Q}_p)$. The map $Q \mapsto \log_E(pQ)/p$ is an isomorphism $E(\mathbb{Q}_p) \to \mathbb{Z}_p$ carrying $E_1(\mathbb{Q}_p)$ onto $p\mathbb{Z}_p$. As $P' \notin E_1(\mathbb{Q}_p)$, $v_p(\log_E(pP')) = 1$, hence $e = 1$. □

**Lemma 4.2.** *$\phi_p(x) \bmod p^2$ depends only on $x \bmod p$. Hence*

$$\lambda(\bar{x}) = \tfrac{1}{2}\, q_p\big(\phi_p(x(P'))\big)$$

*depends only on $E \bmod p^2$ and on the class $\bar{x} \in \widetilde{E}(\mathbb{F}_p) \smallsetminus \{O\}$ of $P'$.*

*Proof.* Since $\phi_p \equiv f(x^p)$, we have $p \mid \phi_p'$, and then $p^2 \mid \phi_p(x + pk) - \phi_p(x)$. The value $\phi_p(x(P'))$ is a unit by the proof of Theorem 4.3. □

**Theorem 4.3.** *Suppose $e \geq 2$ or $p \mid a_1$. Then $q_p(\alpha(P)/\beta(P)) = -\lambda$. In particular, in a model with $p \mid a_1$, $c \geq 2$ if and only if $\lambda = 0$.*

*Proof.* Write $x(P') = a/d'^2$ with $p \nmid d'$. Clearing denominators in $x(P) = \phi_p/\psi_p^2$ and $y(P) = \omega_p/\psi_p^3$, and using (A2) to exclude cancellation, we get $\alpha/\beta = \pm(\phi/\omega)\, d'^{-p^2}$ with $\phi = \phi_p(x(P'))$ and $\omega = \omega_p(P')$; in particular $\phi$ and $\omega$ are units, and

$$q_p(\alpha/\beta) = q_p(\phi) - q_p(\omega).$$

Multiplying the Weierstrass equation at $P$ by $\psi^6$, with $\psi = \psi_p(x(P'))$,

$$\omega^2 + a_1\phi\omega\psi + a_3\omega\psi^3 = \phi^3 + a_2\phi^2\psi^2 + a_4\phi\psi^4 + a_6\psi^6.$$

Here $v_p(\psi) = e$, so all terms of $\omega^2 - \phi^3$ except $a_1\phi\omega\psi$ have valuation at least $2e$, and that term has valuation $v_p(a_1) + e$. Under either hypothesis $\omega^2 \equiv \phi^3 \pmod{p^2}$. Raising to the power $p - 1$ gives $2q_p(\omega) = 3q_p(\phi)$. Hence $q_p(\phi) = 2\lambda$, $q_p(\omega) = 3\lambda$ and $q_p(\alpha/\beta) = -\lambda$. □

*Remark* 4.4. In a general model, $q_p(\alpha/\beta) = -\lambda - \frac{1}{2}a_1\omega(\psi/p)/\phi^2 \bmod p$.

**Theorem 4.5** (Vélu)**.** *Let $K$ be a field of characteristic 0, $m \geq 3$ odd, $D \subset E(\bar{K})$ a Galois-stable cyclic subgroup of order $m$, and $T \in D \smallsetminus \{O\}$ with $x(T) \in K$. Let $\Pi(x) = \prod_{\{\pm Q\} \subset D \smallsetminus \{O\}}(x - x(Q))$ and let $\bar{\phi}/\Pi^2$ be the $x$-coordinate of the Vélu isogeny $E \to E/D$, with $\bar{\phi}$ monic of degree $m$. Then $\phi_m(x(T)) = \bar{\phi}(x(T))^m$.*

*Proof.* Write $[m] = \hat{\pi} \circ \pi$ and $x \circ \hat{\pi} = \phi^\vee/(\psi^\vee)^2$ with $\phi^\vee = \sum_{i \leq m} c_i X^i$, $c_m \neq 0$. Then $\phi_m/\psi_m^2 = N/\mathrm{Den}$ with

$$N = \sum_{i \leq m} c_i\, \bar{\phi}^{\,i}\, \Pi^{2(m-i)}, \qquad \deg N = m^2, \qquad \deg \mathrm{Den} = m^2 - 1.$$

Since $x \circ [m]$ has degree $m^2$, the fraction is reduced, so $N = c_m\phi_m$. At $x(T)$ the factor $\Pi$ vanishes, so $N(x(T)) = c_m\bar{\phi}(x(T))^m$. □

**Corollary 4.6.** *If $E(\mathbb{Q}_p)[p] \neq 0$, then $\lambda(\bar{x}) = 0$ for every $\bar{x} \neq O$.*

*Proof.* Let $D = E(\mathbb{Q}_p)[p]$. For $T \in D \smallsetminus \{O\}$, Theorem 4.5 with $K = \mathbb{Q}_p$ gives $\phi_p(x(T)) = \bar{\phi}(x(T))^p$, a $p$-th power of a unit, so $q_p(\phi_p(x(T))) = 0$. Reduction maps $D \smallsetminus \{O\}$ bijectively onto $\widetilde{E}(\mathbb{F}_p) \smallsetminus \{O\}$, and Lemma 4.2 concludes. □

**Corollary 4.7.** *If $E$ is split at $p$, then $c \geq 2$ in every model.*

*Proof.* Combine Lemma 4.1, Corollary 4.6 and Theorem 4.3. □

**Proposition 4.8.** *Let $p \geq 5$ be a prime of good ordinary reduction with $\#\widetilde{E}(\mathbb{F}_p) = p$. For every $\bar{x} \in \widetilde{E}(\mathbb{F}_p) \smallsetminus \{O\}$,*

$$\lambda(\bar{x}) = 0 \iff E(\mathbb{Q}_p)[p] \neq 0.$$

The implication $\Leftarrow$ is Corollary 4.6. The converse is proved in Appendix A: $\lambda$ is an affine function of the deformation parameter of $\widetilde{E}$ over $\mathbb{Z}/p^2$ with non-zero slope, and exactly one deformation is split. In the language of canonical lifts the equivalence is known [Gro90, LR, DW, OY].

**Theorem 4.9.** *Assume $\#\widetilde{E}(\mathbb{F}_p) = p$, $p \nmid n$ and $E(\mathbb{Q}_p)[p] = 0$. Let $P'$ satisfy* (A2) *and have reduction of order $p$. Then $v_p(h_p(pP')) = 1$ and $v_p(h_p(P')) = -1$.*

*Proof.* Take $p \mid a_1$. Then $e = 1$ by Lemma 4.1 and $b \geq 2$ by Lemma 3.3. By Theorem 4.3 and Proposition 4.8, $q_p(\alpha/\beta) = -\lambda \neq 0$, so $c = 1$. Hence $v_p(h_p(pP')) = 1$, and $h_p(pP') = p^2 h_p(P')$. □

## 5. Proof of Theorem A

Since $p \nmid n$, $v_p(h_p(P)) = v_p(h_p(nP))$, and $nP$ satisfies (A2) for every $P \in E(\mathbb{Q})$.

*Split case.* Let $P \in \Lambda$. If $P \in \Lambda_0$, then $nP$ satisfies (A1) and (A2), and $v_p(h_p(P)) \geq 1$ by Lemma 3.1. If $P \notin \Lambda_0$, put $P' = nP$ and $P'' = pP'$. By Lemma 4.1, $e = v_p(t(P'')) \geq 2$; by Lemma 3.3 and Corollary 4.7, $b \geq 2$ and $c \geq 2$. So $v_p(h_p(P'')) \geq 2$ and $v_p(h_p(P)) = v_p(h_p(P')) \geq 0$. Thus $h_p$ is $\mathbb{Z}_p$-valued on $\Lambda$, and since $p$ is odd so is the pairing. This proves "$\Leftarrow$" in (1), the split half of (2), and, with Theorem 3.2, part (3).

*Non-split case.* By ($\iota$) some $P \in \Lambda$ has reduction of order $p$, and Theorem 4.9 applied to $nP$ gives $v_p(h_p(P)) = -1$. If all entries of $M$ were integral, $h_p$ would be integral on $\Lambda$. This proves "$\Rightarrow$" in (1). For $r = 1$, a generator $Q$ of $\Lambda$ lies outside $\Lambda_0$ and $\mathrm{Reg}_p = h_p(Q)$, which gives the rest of (2). □

## 6. The height matrix modulo $p$

Assume $\#\widetilde{E}(\mathbb{F}_p) = p$, $p \nmid n$ and ($\iota$). Fix a generator $\bar{x}_0$ of $\widetilde{E}(\mathbb{F}_p)$ and let $\chi\colon \Lambda \to \mathbb{F}_p$ be the discrete logarithm of the reduction; so $\ker \chi = \Lambda_0$.

**Proposition 6.1.** *Let $Q \in E(\mathbb{Z}_p)$ have non-zero reduction and let $p \nmid m$. Then*

$$q_p\big(\phi_p(x(mQ))\big) = m^2\, q_p\big(\phi_p(x(Q))\big).$$

*Consequently $\lambda = \lambda_1 \chi^2$ on $\widetilde{E}(\mathbb{F}_p) \smallsetminus \{O\}$, where $\lambda_1 = \lambda(\bar{x}_0)$.*

*Proof.* From the composition rule $\psi_{ab}(R) = \psi_b(R)^{a^2}\psi_a(bR)$ for $(a,b) = (p,m)$ and $(m,p)$, and from $\phi_k(x(R)) = x(kR)\psi_k(R)^2$ with $\phi_m$ monic of degree $m^2$ evaluated at $x(pQ)$, which has negative valuation, one gets

$$\phi_p(x(mQ))\,\psi_m(Q)^{2p^2} = \phi_p(x(Q))^{m^2}(1+\delta), \qquad \delta \in p^2\mathbb{Z}_p.$$

The factor $\psi_m(Q)^{2p^2}$ is a $p$-th power of a unit; apply $q_p$. □

**Lemma 6.2.** *Let $B$ be a symmetric bilinear form on a vector space $V$ over a field of characteristic $\neq 2$, $\chi$ a non-zero linear form, and $v \in V$ with $\chi(v) = 1$.*

(a) *If $B(x,x) = 0 \iff \chi(x) = 0$, then $B(v,v) \neq 0$ and $B = B(v,v)\,\chi \otimes \chi$.*

(b) *If $\chi(x) = 0 \Rightarrow B(x,x) = 0$, then*

$$B(x,y) = B(x,v)\chi(y) + \chi(x)B(y,v) - B(v,v)\chi(x)\chi(y);$$

*in particular* $\operatorname{rank} B \leq 2$.

*Proof.* In both cases polarisation gives $B = 0$ on $\ker\chi \times \ker\chi$. In (a), if $B(w,v) \neq 0$ for some $w \in \ker\chi$, then $x = sw + v$ with $s = -B(v,v)/2B(w,v)$ has $\chi(x) = 1$ and $B(x,x) = 0$, which is excluded. Now decompose $x = (x - \chi(x)v) + \chi(x)v$. □

6.1. **The non-split case.**

**Theorem 6.3.** *Assume moreover $E(\mathbb{Q}_p)[p] = 0$. Then $p\,h_p(P,R) \in \mathbb{Z}_p$ and*

$$p\,h_p(P,R) \equiv 2\lambda_1\,\chi(P)\chi(R) \pmod{p} \qquad (P,R \in \Lambda).$$

*In particular $v_p(h_p(P,R)) = -1$ if and only if $\chi(P)\chi(R) \neq 0$.*

*Proof.* Put $B = p\,h_p$. By Lemma 3.1, $v_p(B(P,P)) \geq 2$ for $P \in \Lambda_0$, and by Theorem 4.9, $v_p(B(P,P)) = 0$ for $P \notin \Lambda_0$. So $B$ is integral and its reduction on $\Lambda/p\Lambda$ satisfies the hypothesis of Lemma 6.2(a). It remains to identify the constant. Take $P' \notin \Lambda_0$ satisfying (A2), a model with $p \mid a_1$, and $P'' = pP'$. Lemma 3.3 gives $v_p(\Lambda_\sigma) \geq 2$, and Theorem 4.3 with (1) gives $A \equiv p\lambda \pmod{p^2}$. So $h_p(P'') \equiv 2p\lambda \pmod{p^2}$, that is, $p\,h_p(P') \equiv 2\lambda(\bar{x}) \pmod{p}$. Proposition 6.1 gives the value $2\lambda_1$ at $\chi = 1$. □

**Corollary 6.4.** *In the situation of Theorem 6.3, take a basis with $\chi(e_1) = 1$ and $e_2, \ldots, e_r \in \Lambda_0$. Then $pM = DA'D$, where $D = \mathrm{diag}(1,p,\ldots,p)$, $A'$ integral and $A'_{11}$ a unit, and*

$$v_p(\mathrm{Reg}_p) = (r-2) + v_p(\det T),$$

*where $T$ is the Schur complement of $A'_{11}$. For $r = 1$ this gives $v_p(\mathrm{Reg}_p) = -1$.*

*Proof.* For $i,j \geq 2$, $p\,h_p(e_i,e_j) \in p^2\mathbb{Z}_p$ by Lemma 3.1, and $p\,h_p(e_1,e_j) \in p\mathbb{Z}_p$ by Theorem 6.3. So $\det(pM) = p^{2(r-1)}\det A'$, while $\det(pM) = p^r\mathrm{Reg}_p$. □

6.2. **The split case.**

**Theorem 6.5.** *Assume moreover $E(\mathbb{Q}_p)[p] \neq 0$, and let $\bar{M} = M \bmod p$. Then $\bar{M} = 0$ on $\ker\chi \times \ker\chi$, so $\rho := \mathrm{rank}\,\bar{M} \leq 2$. In a basis with $\chi(e_1) \neq 0$ and $e_2, \ldots, e_r \in \Lambda_0$,*

$$\bar{M} = \begin{pmatrix} a & b^t \\ b & 0 \end{pmatrix}, \qquad \rho = 2 \iff b \neq 0.$$

*Proof.* $M$ is integral by Theorem A. For $P,R \in \Lambda_0$, Lemma 3.1 puts $h_p(P)$, $h_p(R)$, $h_p(P+R)$ in $2p\mathbb{Z}_p$; apply Lemma 6.2(b). □

**Corollary 6.6.** *In the situation of Theorem 6.5, $v_p(\mathrm{Reg}_p) \geq r - \rho$. If $\rho = 2$, then $v_p(\mathrm{Reg}_p) = (r-2) + v_p(\det T_s)$ for an integral $(r-2)\times(r-2)$ matrix $T_s$. In particular, for $r = 2$, $\mathrm{Reg}_p$ is a unit if and only if $\rho = 2$.*

*Proof.* If $\rho = 2$, change the basis of $\Lambda_0 \otimes \mathbb{Z}_p$ so that $b = (b_1, 0, \ldots, 0)$ with $b_1$ a unit. The upper-left $2 \times 2$ block has determinant $\equiv -b_1^2$, so it is invertible over $\mathbb{Z}_p$, and its Schur complement is $p\,T_s$ with $T_s$ integral. If $\rho \leq 1$, the rows $2, \ldots, r$ of $M$ are divisible by $p$. □

In the split case $q_p(\phi_p(x(P'))) = 0$ by Corollary 4.6, so the second Fermat quotient

$$q^{(2)}(u) = \frac{u^{p-1} - 1}{p^2} \bmod p \qquad (u \in \mathbb{Z}_p^\times,\ q_p(u) = 0)$$

is defined at $u = \phi_p(x(P'))$. It is a homomorphism on its domain.

**Proposition 6.7.** *In the situation of Theorem 6.5, let $\chi(e_1) \neq 0$, $P' = ne_1$, and $\lambda^{(2)} = \frac{1}{2}q^{(2)}(\phi_p(x(P')))$. Then*

$$h_p(e_1) \equiv 2\lambda^{(2)}/n^2 \pmod{p}.$$

*For $r = 1$, $\mathrm{Reg}_p$ is a unit if and only if $\lambda^{(2)} \neq 0$.*

*Proof.* Take $p \mid a_1$ and $P'' = pP'$, so $e \geq 2$. In the Weierstrass identity of Theorem 4.3 the term $a_1\phi\omega\psi$ now has valuation at least 3 and the others at least 4, so $\omega^2 \equiv \phi^3 \pmod{p^3}$. Since $q_p(\phi) = q_p(\omega) = 0$, this gives $2q^{(2)}(\omega) = 3q^{(2)}(\phi)$, hence $q^{(2)}(\alpha/\beta) = -\lambda^{(2)}$; the factor $d'^{p^2}$ has trivial $q^{(2)}$. For $u$ with $q_p(u) = 0$ one has $\log_p u \equiv -p^2 q^{(2)}(u) \pmod{p^3}$, so $A \equiv p^2\lambda^{(2)} \pmod{p^3}$. As $v_p(\Lambda_\sigma) \geq 3$, $h_p(P'') \equiv 2p^2\lambda^{(2)} \pmod{p^3}$, and $h_p(P'') = p^2 n^2 h_p(e_1)$. □

Unlike $\lambda$, the quantity $\lambda^{(2)}$ is not determined by $E \bmod p^2$ and the class of the point: it depends on $E \bmod p^3$ and on the point modulo $p^2$, because $\phi_p'$ is divisible by $p$ but not by $p^2$. So the split value of $v_p(\mathrm{Reg}_p)$ is not a function of the reduction alone. The companion paper [Comp] determines $\bar{M}$ in terms of the first two layers of the filtration of $E(\mathbb{Q}_p)$.

*Remark* 6.8 ($\#\widetilde{E}(\mathbb{F}_5) = 10$). If $p \mid \#\widetilde{E}(\mathbb{F}_p)$ and $p \geq 5$, the Hasse bound gives $\#\widetilde{E}(\mathbb{F}_p) \in \{p, 2p\}$, and $2p$ occurs only for $p = 5$. In that case Theorem 4.9, Theorem 6.3 and Corollary 6.4 remain true with $\chi$ the $p$-part of the reduction, assumed surjective: replace $nP$ by $2nP$ in the proofs.

## 7. CM curves

**Proposition 7.1** ([DW, Prop. 2.1(3)])**.** *If $E$ has complex multiplication and $p \geq 5$ is a prime of good ordinary reduction with $a_p = 1$, then $E(\mathbb{Q}_p)[p] \neq 0$.*

David and Weston prove this for $p > 3$, together with the converse. In outline: $p$ splits in the CM field, so Frobenius lifts to an endomorphism $\pi$ over $\mathbb{Q}_p$ with $\pi + \bar{\pi} = 1$ and $\pi\bar{\pi} = p$; then $\ker(1 - \pi)$ is étale of order $p$, Frobenius acts on it trivially, and its points are therefore rational over $\mathbb{Q}_p$.

**Corollary 7.2.** *Let $E$ have CM, and let $p \geq 5$ be an anomalous prime of good ordinary reduction with $p \nmid n$ and $(\iota)$. Then the height matrix is $p$-integral and $v_p(\mathrm{Reg}_p) \geq \max(r - 2, 0)$. If $r = 2$, then $\mathrm{Reg}_p$ is a unit if and only if $h_p(e_1, e_2) \not\equiv 0 \pmod{p}$ for a basis with $\chi(e_2) = 0$.*

*Proof.* Combine Proposition 7.1, Theorem A and Corollary 6.6. □

Banwait [Ban] proves that for rank-two CM curves, outside an explicit set of primes, the normalised second Taylor coefficient of the Mazur–Tate–Teitelbaum $p$-adic $L$-function is a unit if and only if the cyclotomic regulator is a unit and $\mathrm{Sha}(E/\mathbb{Q})[p^\infty] = 0$. The excluded set contains the anomalous primes. Corollary 7.2 describes the regulator at those primes. Extending the criterion of [Ban] to them would also require the Euler factor $(1 - \alpha^{-1})^2$, which has valuation 2 at an anomalous prime; we do not do this here. For curves with CM by $\mathbb{Z}[i]$ there is at most one anomalous prime [Ban, Lemma 2.2].

## 8. Related work

The negative valuations of $\mathrm{Reg}_p$ at anomalous primes are visible in [MST, §4], for example for $37a$ at $p = 53$. In the normalisation of [Har] the factor $\mathrm{lcm}(\#\widetilde{E}(\mathbb{F}_p), \mathrm{lcm}_\ell c_\ell)$ costs $2v_p(\#\widetilde{E}(\mathbb{F}_p))$ digits of precision at an anomalous prime. Non-anomalousness is used as a hypothesis in [BCS] and [Ban, §2.2]. We have not found Theorem 3.2 stated explicitly.

The condition $E(\mathbb{Q}_p)[p] \neq 0$ is studied as *local torsion* by David and Weston [DW]; Gross's tameness criterion [Gro90] and Ozeki–Yoshida [OY] give it in terms of the canonical lift. It also appears in cryptography: the attacks of Semaev, Smart and Satoh–Araki on anomalous curves [Sem98, Sma99, SA98] fail precisely for lifts congruent to the canonical lift modulo $p^2$ [Vol2]. The elliptic Wieferich conditions of Silverman and Voloch [Sil88, Vol] concern a rational point modulo $p^2$ and are different.

Wuthrich studied $p$-adic heights in families and the valuation of the $p$-adic regulator [Wut04]. We have not found the condition $E(\mathbb{Q}_p)[p] \neq 0$ related to the valuation of $\mathrm{Reg}_p$ in the literature.

## Appendix A. Proof of Proposition 4.8

It remains to show that $\lambda = 0$ implies $E(\mathbb{Q}_p)[p] \neq 0$. Work with a short model $y^2 = x^3 + a_4x + a_6$ and write $\phi_p \equiv f(X)$, $\psi_p \equiv g(X)$ modulo $p$ with $X = x^p$; the roots of $g$ are the $x$-coordinates of $\ker V \smallsetminus \{O\}$.

*Step 1: an Euler identity.* $\phi_p$ is weighted homogeneous of weight $2p^2$ for the weights $4, 6, 2$ of $a_4, a_6, x$. Reducing Euler's identity modulo $p$, and using that $X = x^p$ has weight $2p \equiv 0$, gives $4a_4\,\partial_{a_4}f + 6a_6\,\partial_{a_6}f = 0$ in $\mathbb{F}_p[a_4, a_6][X]$. Put $\tilde{u} = (\partial_{a_6}f)/a_4$.

*Step 2: one slope.* For the deformations $y^2 = x^3 + (a_4 + sp)x + a_6$ and $y^2 = x^3 + a_4x + (a_6 + sp)$, Taylor expansion shows that $\lambda$ is affine in $s$. The two slopes are related by Step 1, since $\lambda$ is invariant under $(x, y) \mapsto (u^2x, u^3y)$; both are multiples of

$$\rho(\bar{x}) = -\tilde{u}(\bar{x})/(4f(\bar{x})),$$

and $f(\bar{x}) \neq 0$ by Vélu's formulas. These deformations exhaust the lifts of $\widetilde{E}$ to $\mathbb{Z}/p^2$, and by Serre–Tate theory exactly one lift is split. As $\lambda$ vanishes on it (Corollary 4.6), it suffices to show $\rho \neq 0$.

*Step 3: all or nothing.* The resultant $\mathrm{Res}_X(\tilde{u}, g)$ is weighted homogeneous of weight $12m$, $m = (p-1)(p-2)(p+3)/12$. Consider $E_t\colon y^2 = x^3 - 3x + (2+t)$ over $\bar{\mathbb{F}}_p((t))$, which has split multiplicative reduction with $v(q) = v(\Delta) = 1$. By the Tate parametrisation [ATAEC, Dok], $\ker V = \langle q \rangle \bmod q^p$; the roots $\beta_j$ of $g$ satisfy $v(\beta_j - 1) = j$ for $1 \leq j \leq (p-1)/2$, and the Newton polygon of $f$ gives $v(f(\beta_j)) = j(p-j)$. Since $\tilde{u} = -\frac{1}{3}\frac{d}{dt}f$, we get $\mathrm{ord}_\Delta \mathrm{Res}_X(\tilde{u}, g) \geq \sum_j (j(p-j) - 1) = m$, and weighted homogeneity forces $\mathrm{Res}_X(\tilde{u}, g) = c\,\Delta^m$ with $c \in \mathbb{F}_p$. So $\rho$ is either non-zero at every ordinary point or identically zero.

*Step 4: $c \neq 0$.* The Frobenius twist of $E_t$ is the member at $t^p$, so $v(d\beta_j/dt) = j - 1$. The two roots of $f$ nearest to $\beta_j$ contribute $1/(1-w) + 1/(1-w^{-1}) = 1$ to $f'(\beta_j)/f(\beta_j)$, which therefore has valuation $-j$ and leading coefficient $p - 2j \equiv -2j$. By the chain rule,

$$12\rho_j = \frac{d}{dt}\log f(\beta_j) - \frac{f'(\beta_j)}{f(\beta_j)}\,\frac{d\beta_j}{dt} = \frac{-j^2 + 2j^2}{t} + O(1).$$

The residue is $j^2 \neq 0$, so $v(\rho_j) = -1$, the bound of Step 3 is an equality, and $c \neq 0$. □

## Appendix B. Lean formalisation

The Lean project for this paper has eleven files, each importing only Mathlib (Lean 4 `v4.33.1`, Mathlib `0df444a3`). They contain no `sorry` and no `native_decide`, and all `#print axioms` lines are subsets of {`propext`, `Classical.choice`, `Quot.sound`}. Sources and axiom logs are in [Repo]. File $k$ is `PadicHeightValuation`$k$`.lean` (no suffix for $k = 1$).

| Result | Lean declarations | File |
|---|---|---|
| Thm. 3.2, Lemma 3.3 | `norm_det_div_sq_le`, `fermatQuot_additive` | 1 |
| Lemma 4.2 | `eval_sub_eval_dvd_sq` | 2 |
| Thm. 4.3 | `fermatQuot_sub_iff_of_sq_eq_cube`, `sq_sub_cube_dvd_of_dvd_a1`, `dvd_three_mul_sub_two_mul` | 2, 8 |
| Cor. 4.6, 4.7 | `sq_dvd_pow_sub_one_of_isPow`, `sq_dvd_alpha_sub_beta_of_isPow` | 3 |
| Prop. 6.1 | `fermatQuot_scale`, `fermatQuot_scale_zero_iff` | 4 |
| Theorem A | `regulator_valuation_of_split_anomalous`, `regulator_valuation_of_split_anomalous_iff_split`, `quadratic_value_mem_of_entries_mem` | 5, 8 |

| | | |
|---|---|---|
| App. A, Step 2 | `slope_of_rho`, `fermatQuot_deform`, `affine_zero_unique`, `lambda_eq_zero_iff_split` | 5 |
| App. A, Step 3 | `weight_split`, `sum_pole_orders`, `cusp_sum_eq_m`, `eq_C_mul_of_dvd_of_natDegree_eq` | 6 |
| App. A, Step 4 | `first_term_residue`, `second_term_leading_ne_zero`, `residues_do_not_cancel` | 7 |
| Thm. 4.9 | `exists_shift_dvd_a1`, `add_val_exactly_one`, `not_dvd_of_val_one` | 8 |
| Lemma 6.2(a), Thm. 6.3, Cor. 6.4 | `bilin_rank_one_of_zero_iff`, `det_conj_diagonal_one_p`, `reg_eq_of_det_eq` | 9 |
| Lemma 6.2(b), Thm. 6.5, Cor. 6.6, Prop. 6.7 | `bilin_eq_of_vanishing_on_hyperplane`, `gram_det_three_eq_zero`, `det_two_dvd_iff`, `pow_dvd_det_of_offdiag_block`, `fermatQuot2_add`, `split_const_eq` | 10 |
| Remark on $\#\widetilde{E}(\mathbb{F}_5) = 10$ | `card_cases`, `two_p_forces_five` | 22 |

## Use of AI

This research was carried out with the assistance of Claude Code (Anthropic). The constructions, proofs, computations and the Lean formalisation were developed with the AI system under the author's direction, and separate sessions of the system re-checked the Lean development and the computations. The author reviewed the arguments, checked the Lean development and its axiom logs, and takes full responsibility for the content.

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